Mostafa Khalifa

FULL-STACK SOFTWARE ENGINEER

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Mostafa Khalifa Mostafa-Khalifaa **Military Status:** Exempted

Summary

Software Engineer, Computer Science graduate (Excellent with Honors), and ITI Open Source Track graduate. Experienced in building full-stack web applications, RESTful APIs, and containerized microservices using Python (FastAPI, Django), React, TypeScript, and Docker. Focused on clean architecture, scalable data-driven services, and cloud integration.

Education

Information Technology Institute (ITI) **Oct 2025 – Aug 2026**

9-Month Professional Diploma, Open Source Track *Cairo, Egypt*

Pharos University **Sep 2021 – Jul 2025**

Bachelor of Science in Computer Science – GPA: 3.6/4.0 (Excellent) *Alexandria, Egypt*

- **Graduation Project: Solving Sphere** (Grade: A+) | Code
- Built a competitive programming platform using Laravel and React (Inertia.js) featuring role-based access, study roadmaps, and community blogs.
- Developed a Python/FastAPI microservice with Selenium to scrape live HackerEarth data for real-time code evaluation and user profiles.
- Integrated Ollama API for AI-driven debugging hints and established CI/CD pipelines using GitHub Actions and Pest testing.

Internships

Almadrasa **Jul 2024 – Dec 2024**

Front-End Web Development Internship *Online*

- Engineered and deployed 20+ interactive web projects, establishing a strong foundation in JavaScript, HTML5, CSS3, and Python.
- Managed the full frontend lifecycle from design handoff to deployment, focusing on responsive UIs and modern React applications.

RMZtech **Jul 2023 – Aug 2023**

Front-End Development Intern *Alexandria, Egypt*

- Built responsive UIs using Vue.js, Tailwind CSS, and Bootstrap during internship training.
- Followed accessibility standards to ensure cross-device usability.
- Used GitHub for collaborative version control and issue tracking.
- Enhanced front-end modularity by introducing reusable UI components.

Projects

Manikan AI-Powered SaaS Platform | *Python (FastAPI), LangGraph, Next.js, PostgreSQL* **Code** | **Live**

- Engineered a microservices-based SaaS platform featuring backend AI services built with Python and FastAPI.
- Implemented an AI Agent using Retrieval-Augmented Generation (RAG) and LangGraph to process data and generate dynamic recommendations.
- Developed responsive frontend interfaces using Next.js and React, seamlessly integrating complex machine learning APIs.

ITI-Attendance & Grading Platform | *Vue 3, TypeScript, Laravel 11, PostgreSQL, Docker* **Code** | **Live**

- Collaborated within a 5-member team to engineer an enterprise-grade academic management system, handling complex hierarchical data and operational workflows.
- Developed a robust Laravel REST API with strict Role-Based Access Control (RBAC) via Sanctum, utilizing PostgreSQL and Docker.
- Built a modular SPA frontend using Vue 3 and TypeScript, featuring QR-code attendance tracking and real-time analytics dashboards.

Fundly Crowdfunding | *React 19, TypeScript, Tailwind CSS, Zustand, Python (Django)* **Code** | **Live**

- Built a modern crowdfunding web application, contributing to both UI implementation and backend API integration.
- Integrated a Python (Django REST Framework) backend with interactive community features including threaded comments and real-time donation tracking.
- Implemented protected routing and JWT authentication, ensuring a highly responsive user interface deployed on Vercel.

Skills

Languages: JavaScript, TypeScript, Python, PHP, HTML5, CSS3, SQL

Front-End: React (Vite), Vue.js, Tailwind CSS, Zustand, RxJS

Back-End: Node.js (Express), Django, Laravel, FastAPI

AI & Data Science: LangChain, RAG, Agentic AI, Pandas, NumPy, Scikit-Learn, Matplotlib

Databases: MySQL, PostgreSQL

Tools & Cloud: Git, GitHub, Linux, Docker, n8n, Postman, Figma, Vercel, Power BI, AWS

Concepts: RESTful APIs, JWT Auth, MVC Architecture, Data Structures, OOP, Agile

Soft Skills: Problem Solving, Effective Communication, Teamwork, Adaptability, Leadership

Leadership / Extracurricular

International Collegiate Programming Contest (ICPC)

Participant & Student Mentor

- Participated in regional contests, focusing on software engineering fundamentals and clean architectural patterns.
- Mentored junior student teams by simplifying core programming logic and collaborative development practices.

Student Union – Pharos University

Head of Committee

- Led a student committee for two consecutive years, managing campus-wide events, academic initiatives, and student engagement programs.
- Directed cross-functional collaboration with other committees to enhance student experience and community impact.

Languages

Arabic: Native Proficiency

English: Very Good Reading and Writing Proficiency